Chapter 1

Supplements 0

Remark 1.1 (Physics notation: fields, action and propagator)**.** In the components of the physics notation key after Section 1.2:

- x^m, $m = 1, \dots, 4$, are the coordinates of Euclidean $\mathbb{R}^4$; $W_{\mathbb{R}}^{\perp}$ carries the $SO(6)_R$ vector index $I = 1, \dots, 6$ and W, the $\mathbf{4}$ of $SU(4)_R$, the index $i = 1, \dots, 4$.
- $A|_V = A_m$ is the gauge field, $\iota_w A = w_I A_I$ the six scalars, $\lambda = (\lambda_\alpha{}^i, \bar\lambda_{\dot\alpha i})$ the four gauginos, c the Faddeev–Popov ghost, and A^*_m, A^*_I, $\lambda^{*\alpha}{}_i$, $\bar\lambda^{*\dot\alpha i}$, c^* the antifields, all $\mathfrak{g}$-valued.
- S is the BV action of Euclidean $\mathcal{N} = 4$ super Yang–Mills theory with weight $e^{S/\hbar}$, Q its linearised BRST operator, I the interaction and $\{-,-\}$ the BV antibracket.
- Q^{GF} is the Feynman-gauge gauge-fixing operator, $H = -\partial_m\partial_m$ ($|p|^2$ in momentum space), K_t the heat kernel with momentum factor $e^{-t|p|^2}$, and $P(\varepsilon,\infty)$ the Feynman-gauge propagator with proper-time cutoff ε (mass dimension -2): each component is its fibre factor times $\int_\varepsilon^\infty e^{-t|p|^2}dt = e^{-\varepsilon|p|^2}/|p|^2$. For $\mathfrak{su}(N)$, with $A_M = -igA^a_M T^a$, $M = (m, I)$, and $\lambda = -\frac{ig}{\sqrt{2}}\lambda^a T^a$, the edge $\hbar P(\varepsilon,\infty)$ of $\hbar\partial_{P(\varepsilon,\infty)}$ has the components

$$A^a_M \otimes A^b_N :\ \delta^{ab}\delta_{MN}\,\frac{e^{-\varepsilon|p|^2}}{|p|^2}, \qquad \lambda^{a\,i}_\alpha \otimes \bar\lambda^b_{\dot\beta j} :\ -\delta^{ab}p_m\sigma^m_{\alpha\dot\beta}\delta^i_j\,\frac{e^{-\varepsilon|p|^2}}{|p|^2}.$$

- $W(P,J)$ is $\hbar$ times the sum of connected Feynman graphs with edges $\hbar P$ and vertices $J/\hbar$, each graph weighted by $\hbar$ to its number of loops; ε^{-1} and $\log\varepsilon$ are the quadratic and logarithmic divergences.
- With $\omega = f\,d^4x$, $\int\omega\,(\iota_{w_1}A, \iota_{w_2}A) = w_{1I}w_{2J}\int d^4x\,f\,(A_I, A_J)$, equal to $2\hbar\,w_{1I}w_{2J}\int d^4x\,f\,A^a_I A^a_J$ for $\mathfrak{su}(N)$; δ is its source, $O(\varepsilon)$ the bare operator including its counterterm, and $O[\infty]$ the renormalised operator.
- $\Delta = 2E = 2x^m\partial_m$, and $[(d_{CE}+Q)O[\infty]](x_1)$ is the quantum-corrected transformation of O along the generator x_1 of the key: P_m, K_m, Δ, rotations, R-symmetry, Q^i_α, $\bar Q_{\dot\alpha i}$, $S^\alpha{}_i$, $\bar S^{\dot\alpha i}$.

Remark 1.2 (Physics notation: the UV boundary)**.**

- K_0 is $-\delta^4(x-y)$ times the inverse fibre pairing on each field–antifield pair; K_t has $\delta^4(x-y)$ replaced by $(4\pi t)^{-2}e^{-|x-y|^2/4t}$, momentum factor $e^{-t|p|^2}$, so $K_\infty = 0$ and $K_\varepsilon - K_0$ carries the factor $e^{-\varepsilon|p|^2} - 1$.
- $L_t = (Q^{GF}\otimes 1)K_t$ is the integrand of the proper-time representation $P(\varepsilon,\infty) = \int_\varepsilon^\infty L_t\,dt$: the fibre factor of the propagator times $e^{-t|p|^2}$.

- $P(0,\infty)$ is the Feynman-gauge propagator without cutoff, fibre factor times $1/|p|^2$, and $Q\,P(0,\infty)\star + P(0,\infty)\star Q = 1$ makes $P(0,\infty)\star$ a Green's operator of the free BRST operator Q.
- $\Delta_\varepsilon = -\partial_{K_\varepsilon}$ is the BV Laplacian regularised by the heat kernel at proper time ε; the quantum master equation at the UV boundary reads $(d_{CE}+Q)J + \frac{1}{2}\{J,J\}_\varepsilon + \hbar\Delta_\varepsilon J = 0$, with classical part the Zinn-Justin equation.
- The ε^{-1} mass terms are $(A_m, A_m) + (A_I, A_I)$, and the finite counterterm forced by the master equation is $(\partial_m A_m, \partial_n A_n)$ (Section 0.4).

Remark 1.3 (Physics notation: the objects of the chapters)**.** The Feynman-gauge operator of Chapter 3 (Definition **??**) is, with the slot of the image in brackets,

$$A^*_M \mapsto -2A^*_M\ [A_M], \qquad A_m \mapsto \partial_m A_m\ [c], \qquad c^* \mapsto \partial_m c^*\ [A^*_m],$$
$$\lambda^{*\alpha}{}_i \mapsto -i\,\partial_m \lambda^{*\alpha}{}_i \sigma^m_{\alpha\dot\alpha}\ [\bar\lambda_{\dot\alpha i}], \qquad \bar\lambda^{*\dot\alpha i} \mapsto i\,\sigma^m_{\alpha\dot\alpha}\partial_m \bar\lambda^{*\dot\alpha i}\ [\lambda_\alpha{}^i],$$

with $H = [Q, Q^{GF}] = -\partial_m\partial_m$. The single letters are $\int\omega\,(X, A_I)$, $\int\omega\,(X, v_m A_m)$, $\int\omega\,(\zeta,\lambda)$, $\int\omega\,(X,c)$; the scalar bilinears are $w_{1I}w_{2J}\int\omega\,(A_I,A_J)$, whose traceless part is the $\mathbf{20'}$ of $SU(4)_R$ and whose trace $\int\omega\,(A_I,A_I)$ is the Konishi operator. The other superconformal components of Chapter 7 are those along Q^i_α, $\bar Q_{\dot\alpha i}$, $S^\alpha{}_i$, $\bar S^{\dot\alpha i}$ and K_m.

Remark 1.4 (Physics notation: the results of Step 1)**.** With $(A_I,A_I) = \frac{1}{4}(A^{ij}, A_{ij})$, $F^0_{mn} = \partial_m A_n - \partial_n A_m$ and $d^*A|_V = -\partial_m A_m$, the quadratic counterterm of Chapter 4 is

$$\frac{\hbar\,\mathrm{tr}_{\mathfrak g}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\int d^4x\Big\{\frac{9}{4\varepsilon}\big[(A_m,A_m)+(A_I,A_I)\big] + \log\varepsilon\Big[-\tfrac{1}{2}(F^0_{mn},F^0_{mn}) - (\partial_m A_I, \partial_m A_I) + 8i\big(\bar\lambda_{\dot\alpha i}, \bar\sigma^{m\dot\alpha\alpha}\partial_m\lambda_\alpha{}^i\big) - (\partial_m c, A^*_m)\Big]\Big\},$$

gauge-boson and scalar mass terms at order ε^{-1} and kinetic terms at order $\log\varepsilon$. The $\log\varepsilon$ part through quartic order is the unit times $\log\varepsilon\,(Q+\{I,-\})$ of

$$\int d^4x\Big\{2(A_m,A^*_m) + 2(A_I,A^*_I) - 4\big[(\lambda_\alpha{}^i, \lambda^{*\alpha}{}_i) - (\bar\lambda_{\dot\alpha i}, \bar\lambda^{*\dot\alpha i})\big] - (c^*, c)\Big\},$$

the BV form of a wave-function renormalisation, with vanishing one-loop beta function of g. The finite counterterm is $-\frac{27}{16}$ units times $\int d^4x\,(\partial_m A_m, \partial_n A_n)$, removing the obstruction $-\frac{27}{8}$ units times $\int d^4x\,(\partial^2 c, \partial_m A_m)$. In Chapters 5 and 6, $(\zeta,\lambda) = (\lambda_\alpha{}^i, \zeta^\alpha{}_i) - (\bar\lambda_{\dot\alpha i}, \bar\zeta^{\dot\alpha i})$, $(w_1,w_2) = w_{1I}w_{2I}$, the traceless bilinear is the $\mathbf{20'}$ and the trace $\int\omega\,(A_I,A_I) = -\frac{4\hbar}{g^2}\int\omega\,\mathrm{tr}(A_I A_I)$ the Konishi operator. The coefficient times the unit is the one-loop anomalous dimension γ_O, the scaling dimension being $w_O + \gamma_O$; for $\mathfrak{su}(N)$ the table gives $\gamma = 0$ for the $\mathbf{20'}$, $\frac{3g^2N}{4\pi^2}$ for the Konishi operator and, in Feynman gauge, $\frac{g^2N}{8\pi^2}$ for A_I and A_m, $\frac{g^2N}{4\pi^2}$ for λ, $\frac{g^2N}{16\pi^2}$ for c.

Chapter 2

Supplements 1

Remark 2.1 (Physics notation: Clifford relation, Gamma-pairing and Dirac operator)**.** In the two-component notation of the physics notation key after Section 1.2, the Clifford relation along V and along $W_{\mathbb{R}}^{\perp}$ reads

$$\sigma^m\bar\sigma^n+\sigma^n\bar\sigma^m=-2\delta^{mn},\qquad u^{ik}w_{kj}+w^{ik}u_{kj}=-2(u,w)\,\delta^i_j\quad(u,w\in W_{\mathbb{R}}^{\perp}),$$

and the Γ-pairing on Σ and on Σ^* has the components

$$\begin{aligned}
&\Gamma(\xi_1,\xi_2)\big|_V^m=i\big(\bar\xi_{2\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\xi_{1\alpha}{}^i+\bar\xi_{1\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\xi_{2\alpha}{}^i\big),\\
&\Gamma(\xi_1,\xi_2)_{kl}=2\varepsilon_{jikl}\xi_{1\alpha}{}^j\varepsilon^{\alpha\beta}\xi_{2\beta}{}^i-2\big(\bar\xi_{1\dot\alpha k}\varepsilon^{\dot\alpha\dot\beta}\bar\xi_{2\dot\beta l}-\bar\xi_{1\dot\alpha l}\varepsilon^{\dot\alpha\dot\beta}\bar\xi_{2\dot\beta k}\big),\\
&\Gamma(\eta_1,\eta_2)\big|_V^m=-i\big(\eta_2{}^\alpha{}_i\sigma^m_{\alpha\dot\alpha}\bar\eta_1^{\dot\alpha i}+\eta_1{}^\alpha{}_i\sigma^m_{\alpha\dot\alpha}\bar\eta_2^{\dot\alpha i}\big),\\
&\Gamma(\eta_1,\eta_2)^{kl}=-2\varepsilon^{klpj}\varepsilon_{\alpha\beta}\eta_1{}^\beta{}_j\eta_2{}^\alpha{}_p+2\big(\varepsilon_{\dot\beta\dot\alpha}\bar\eta_1^{\dot\alpha l}\bar\eta_2^{\dot\beta k}-\varepsilon_{\dot\beta\dot\alpha}\bar\eta_1^{\dot\alpha k}\bar\eta_2^{\dot\beta l}\big),
\end{aligned}$$

with $\Gamma_{kl}=\Gamma^I\Sigma^I_{kl}$, $\Gamma^{kl}=\frac12\varepsilon^{klij}\Gamma_{ij}$ and $(\Gamma,u)=\Gamma^m u_m+\frac14\Gamma_{ij}u^{ij}$. On two-forms along V,

$$\rho(q(e_m\wedge e_n))\colon\quad \xi_\alpha{}^i\mapsto-\tfrac12(\sigma^m\bar\sigma^n-\sigma^n\bar\sigma^m)_\alpha{}^\beta\xi_\beta{}^i,\qquad \bar\xi_{\dot\alpha i}\mapsto\tfrac12\bar\xi_{\dot\beta i}(\bar\sigma^m\sigma^n-\bar\sigma^n\sigma^m)^{\dot\beta}{}_{\dot\alpha}.$$

Further $d=dx^m\partial_m$, $\mathrm{dvol}=d^4x$, $(d^*dA)_n=-\partial_m\partial_mA_n+\partial_n\partial_mA_m$ on one-forms, and the Dirac operators are

$$\begin{aligned}
&(\not\partial\xi)^\alpha{}_i=i\,\partial_m\bar\xi_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha},\qquad (\not\partial\xi)^{\dot\alpha i}=-i\,\bar\sigma^{m\dot\alpha\alpha}\partial_m\xi_\alpha{}^i,\\
&(\not\partial|_{\Sigma^*}\eta)_\alpha{}^i=-i\,\sigma^m_{\alpha\dot\alpha}\partial_m\bar\eta^{\dot\alpha i},\qquad (\not\partial|_{\Sigma^*}\eta)_{\dot\alpha i}=i\,\partial_m\eta^\alpha{}_i\sigma^m_{\alpha\dot\alpha}.
\end{aligned}$$

Remark 2.2 (Physics notation: four-dimensional chirality)**.** $\mathrm{Spin}(4,\mathbb{C})=SL(2,\mathbb{C})\times SL(2,\mathbb{C})$ is the complexification of $SU(2)_L\times SU(2)_R$, and $\mathrm{Spin}(6,\mathbb{C})\cong SL(4,\mathbb{C})$ that of $SU(4)_R$, with $W=S^6_+$ the $\mathbf{4}$. The eigenspaces of Proposition **??** are

$$+1:\ \ \xi_\alpha{}^i\in S_+\otimes W,\quad \eta^\alpha{}_i\in S_+\otimes W^*;\qquad -1:\ \ \bar\xi_{\dot\alpha i}\in S_-\otimes W^*,\quad \bar\eta^{\dot\alpha i}\in S_-\otimes W.$$

On the Dirac column $(\xi_\alpha{}^i,\bar\eta^{\dot\alpha i})$ at fixed i,

$$\rho(e_m)=\begin{pmatrix}0&-i\sigma^m\\-i\bar\sigma^m&0\end{pmatrix},\qquad \rho(e_1)\rho(e_2)\rho(e_3)\rho(e_4)=-\gamma_{5S}=\gamma_{5W}=\mathrm{diag}(1,1,-1,-1),$$

so the eigenvalue $+1$ is Srednicki's left-handed chirality and Weinberg's $\gamma_5=+1$. In indices, $\rho(v)$ exchanges undotted and dotted indices and keeps the position of the $SU(4)$ index, $\xi_\alpha{}^i\mapsto(\rho(v)\xi)^{\dot\alpha i}$; $\rho(w)$ keeps the spinor type and maps the $\mathbf{4}$ to the $\bar{\mathbf{4}}$ through w_{ji}, $\xi_\alpha{}^i\mapsto(\rho(w)\xi)^\alpha{}_j$, and the $\bar{\mathbf{4}}$ to the $\mathbf{4}$ through w^{ji}. The opposite orientation exchanges dotted and undotted indices in all four eigenspaces.

Remark 2.3 (Physics notation: components of the fields and antifields)**.** Under $SU(2)_L \times SU(2)_R \times SU(4)_R$:

- c: the Faddeev–Popov ghost, a singlet.
- A: $A_M = (A_m, A_I)$, the gauge field $A_m \in (\mathbf{2},\mathbf{2},\mathbf{1})$ and the six scalars $A_I \in (\mathbf{1},\mathbf{1},\mathbf{6})$, with $A_{ij} = A_I \Sigma^I_{ij}$ and $A^{ij} = \frac{1}{2}\varepsilon^{ijkl}A_{kl}$.
- λ: ${\lambda_\alpha}^i \in (\mathbf{2},\mathbf{1},\mathbf{4})$ and $\bar\lambda_{\dot\alpha i} \in (\mathbf{1},\mathbf{2},\bar{\mathbf{4}})$.
- Antifields: $A^*_M = (A^*_m, A^*_I)$ with $A^*_{ij} = A^*_I \Sigma^I_{ij}$, $A^{*ij} = \frac{1}{2}\varepsilon^{ijkl}A^*_{kl}$; ${\lambda^*}^\alpha{}_i \in (\mathbf{2},\mathbf{1},\bar{\mathbf{4}})$ and $\bar\lambda^{*\dot\alpha i} \in (\mathbf{1},\mathbf{2},\mathbf{4})$; c^*, a singlet.

The identifications of duals give the fibre pairings, with $(-,-)$ on $\mathfrak{g}$ kept on the colour factors,

$$(A, A^*) = (A_m, A^*_m) + (A_I, A^*_I) = (A_m, A^*_m) + \tfrac{1}{4}(A^{ij}, A^*_{ij}),$$
$$(\lambda, \lambda^*) = ({\lambda_\alpha}^i, {\lambda^*}^\alpha{}_i) - (\bar\lambda_{\dot\alpha i}, \bar\lambda^{*\dot\alpha i}), \qquad (c^*, c).$$

Remark 2.4 (Physics notation: the BV symplectic form)**.** With $\mathrm{dvol} = d^4x$,

$$\langle A, A^*\rangle = \int d^4x \left[(A_m, A^*_m) + (A_I, A^*_I)\right], \qquad \langle c^*, c\rangle = \int d^4x\, (c^*, c),$$
$$\langle \lambda^*, \lambda\rangle = \int d^4x \left[({\lambda_\alpha}^i, {\lambda^*}^\alpha{}_i) - (\bar\lambda_{\dot\alpha i}, \bar\lambda^{*\dot\alpha i})\right],$$

and $\langle A^*, A\rangle$, $\langle \lambda, \lambda^*\rangle$, $\langle c, c^*\rangle$ are their negatives. The field–antifield pairs are (A_m, A^*_m), (A_I, A^*_I), $({\lambda_\alpha}^i, {\lambda^*}^\alpha{}_i)$, $(\bar\lambda_{\dot\alpha i}, \bar\lambda^{*\dot\alpha i})$, the last with the sign -1, and (c, c^*); the even members A_M, ${\lambda^*}^\alpha{}_i$, $\bar\lambda^{*\dot\alpha i}$, c^* are the Grassmann-even coordinates.

Remark 2.5 (Physics notation: kernel of the identity, antibracket and BV Laplacian)**.** With $\delta(x-y) = \delta^4(x-y)$, the components of K_0 are $-\delta(x-y)\,\delta_{MN}(-,-)^{-1}$ on $A_M \otimes A^*_N$ and $A^*_N \otimes A_M$, $-\delta(x-y)\,(-,-)^{-1}$ on $c \otimes c^*$ and $c^* \otimes c$, and on $\lambda \otimes \lambda^*$ and $\lambda^* \otimes \lambda$ $-\delta(x-y)\,(-,-)^{-1}$ times the canonical element of $\Sigma \otimes \Sigma^*$, whose components are $\delta^\beta_\alpha \delta^i_j$ between ${\lambda_\alpha}^i$ and ${\lambda^*}^\beta{}_j$ and $-\delta^{\dot\beta}_{\dot\alpha}\delta^j_i$ between $\bar\lambda_{\dot\alpha i}$ and $\bar\lambda^{*\dot\beta j}$. Further, $\{-,-\}$ is the BV antibracket and $\Delta_0 = -\partial_{K_0}$ the BV Laplacian, which contracts a field with its antifield at coincident points.

Remark 2.6 (Physics notation: the action in components)**.** With $D_m = \partial_m + [A_m, -]$, the components of $F_A = dA + \frac{1}{2}[A \wedge A]$ are

$$F_{mn} = \partial_m A_n - \partial_n A_m + [A_m, A_n], \qquad (F_A)_{mI} = D_m A_I, \qquad (F_A)_{IJ} = [A_I, A_J],$$

and $(F_A, F_A) = \sum_{M<N}(F_{MN}, F_{MN})$, so that

$$-\tfrac{1}{4}(F_A, F_A) = -\tfrac{1}{8}(F_{mn}, F_{mn}) - \tfrac{1}{4}(D_m A_I, D_m A_I) - \tfrac{1}{8}([A_I, A_J], [A_I, A_J]),$$

summed over m, n, I, J. The density of (**??**) is

$$\begin{aligned} S = \int d^4x \Big\{ & -\tfrac{1}{8}(F_{mn}, F_{mn}) - \tfrac{1}{16}(D_m A^{ij}, D_m A_{ij}) - \tfrac{1}{128}\big([A^{ij}, A^{kl}], [A_{ij}, A_{kl}]\big) \\ & + \tfrac{i}{2}\big({\lambda_\alpha}^i, \bar\sigma^{m\dot\alpha\alpha} D_m \bar\lambda_{\dot\alpha i}\big) + \tfrac{i}{2}\big(\bar\lambda_{\dot\alpha i}, \bar\sigma^{m\dot\alpha\alpha} D_m {\lambda_\alpha}^i\big) \\ & + \tfrac{1}{2}\big({\lambda_\alpha}^j, \varepsilon^{\alpha\beta}[A_{ji}, {\lambda_\beta}^i]\big) - \tfrac{1}{2}\big(\bar\lambda_{\dot\alpha j}, \varepsilon^{\dot\alpha\dot\beta}[A^{ji}, \bar\lambda_{\dot\beta i}]\big) \\ & + (D_m c, A^*_m) + \tfrac{1}{4}([A^{ij}, c], A^*_{ij}) + ([c, {\lambda_\alpha}^i], {\lambda^*}^\alpha{}_i) - ([c, \bar\lambda_{\dot\alpha i}], \bar\lambda^{*\dot\alpha i}) + \tfrac{1}{2}([c, c], c^*)\Big\}. \end{aligned}$$

For $\mathfrak{su}(N)$ in the normalisation of the key, with $[T^a, T^b] = i f^{abc} T^c$, $A^a_{ij} = A^a_I \Sigma^I_{ij}$, $A^{a\,ij} = \frac{1}{2}\varepsilon^{ijkl}A^a_{kl}$ and Srednicki's contractions, the scalar potential and the Yukawa terms of $S/\hbar$ are

$$-\tfrac{g^2}{4} f^{abe} f^{cde} A^a_I A^b_J A^c_I A^d_J - \tfrac{g}{2} f^{abc} A^b_{ji}\big(\lambda^{aj}\lambda^{ci}\big) - \tfrac{g}{2} f^{abc} A^{b\,ji}\big(\bar\lambda^a_j \bar\lambda^c_i\big).$$

Remark 2.7 (Physics notation: free part, Fourier convention and real slice)**.** Up to total derivatives, (**??**) is

$$\tfrac{1}{2}\langle e, Qe\rangle = \int d^4x \Big\{ -\tfrac{1}{8}(F^0_{mn}, F^0_{mn}) - \tfrac{1}{4}(\partial_m A_I, \partial_m A_I) + i\big(\bar\lambda_{\dot\alpha i}, \bar\sigma^{m\dot\alpha\alpha}\partial_m \lambda_\alpha{}^i\big) + (\partial_m c, A^*_m)\Big\},$$

with $F^0_{mn} = \partial_m A_n - \partial_n A_m$. The coefficient $-\frac{1}{4}$ of Proposition **??** gives $-\frac{1}{8}(F_{mn}, F_{mn})$, which for $\mathfrak{su}(N)$ is $-\frac{1}{4}F^a_{mn}F^a_{mn}$ in $S/\hbar$; $S^{(1)}(Q_1)$ is the supersymmetry term of the equivariant action, in components at the end of Section 2.3. The Fourier convention is $e^{i(p,x)} = e^{ip_m x_m}$, so $\partial_m \mapsto ip_m$ and $|p|^2 = p_m p_m$. On the real slice of Remark **??** the components A^a_m, A^a_I of $A_M = -igA^a_M T^a$ are real and the complex conjugate of A^a_{ij} is $A^{a\,ij}$; the eigenvalue $-\frac{1}{2}|p|^2$ belongs to the three transverse components of A_m and the six A_I, the eigenvalue 0 to $A_m \propto p_m$.

Remark 2.8 (Physics notation: the linearised BRST operator)**.** With the slot of the image in brackets and $\partial^2 = \partial_m\partial_m$,

$$c \mapsto -\partial_m c\ [A_m], \quad A_m \mapsto \tfrac{1}{2}(\partial^2 A_m - \partial_m\partial_n A_n)\ [A^*_m], \quad A_I \mapsto \tfrac{1}{2}\partial^2 A_I\ [A^*_I], \quad A^*_m \mapsto -\partial_m A^*_m\ [c^*],$$

$$\lambda_\alpha{}^i \mapsto -i\,\bar\sigma^{m\dot\alpha\alpha}\partial_m\lambda_\alpha{}^i\ [\bar\lambda^{*\dot\alpha i}], \qquad \bar\lambda_{\dot\alpha i} \mapsto i\,\partial_m\bar\lambda_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\ [\lambda^{*\alpha}{}_i],$$

in momentum space with $\partial_m \mapsto ip_m$: the linearised gauge transformation $-\partial_m c$, the Maxwell and Klein–Gordon operators of A_m and A_I, the Weyl operators of $\lambda_\alpha{}^i$ and $\bar\lambda_{\dot\alpha i}$, and the divergence $-\partial_m A^*_m$ in the c^* slot.

Remark 2.9 (Physics notation: the vertices in components)**.** Summed over m, n, I, J, the terms of (**??**) are

$$-\tfrac{1}{2}\big(dA, \tfrac{1}{2}[A\wedge A]\big) = -\tfrac{1}{2}(\partial_m A_n, [A_m, A_n]) - \tfrac{1}{2}(\partial_m A_I, [A_m, A_I]),$$

$$-\tfrac{1}{4}\big(\tfrac{1}{2}[A\wedge A], \tfrac{1}{2}[A\wedge A]\big) = -\tfrac{1}{8}([A_m, A_n], [A_m, A_n]) - \tfrac{1}{4}([A_m, A_I], [A_m, A_I]) - \tfrac{1}{8}([A_I, A_J], [A_I, A_J]),$$

$$([A, c], A^*) = ([A_m, c], A^*_m) + ([A_I, c], A^*_I), \qquad ([c, \lambda], \lambda^*) = ([c, \lambda_\alpha{}^i], \lambda^{*\alpha}{}_i) - ([c, \bar\lambda_{\dot\alpha i}], \bar\lambda^{*\dot\alpha i}).$$

In the vertex functions $\iota_p A(p_j) = p_m A_m(p_j)$, $[A(p_1), A(p_2)] = [A_n(p_1), A_n(p_2)] + [A_I(p_1), A_I(p_2)]$, and the $\lambda\lambda A$ vertex is the gauge coupling plus the Yukawa coupling,

$$\begin{aligned}\big(\lambda(p_1), \rho([A(p_3), \lambda(p_2)])\big) = {}& i\big(\lambda_\alpha{}^i(p_1), \bar\sigma^{m\dot\alpha\alpha}[A_m(p_3), \bar\lambda_{\dot\alpha i}(p_2)]\big) + i\big(\bar\lambda_{\dot\alpha i}(p_1), \bar\sigma^{m\dot\alpha\alpha}[A_m(p_3), \lambda_\alpha{}^i(p_2)]\big)\\ &+ \big(\lambda_\alpha{}^j(p_1), \varepsilon^{\alpha\beta}[A_{ji}(p_3), \lambda_\beta{}^i(p_2)]\big) - \big(\bar\lambda_{\dot\alpha j}(p_1), \varepsilon^{\dot\alpha\dot\beta}[A^{ji}(p_3), \bar\lambda_{\dot\beta i}(p_2)]\big).\end{aligned}$$

The r_j are Grassmann-odd sources for the legs λ, c and A^*.

Remark 2.10 (Physics notation: gauge transformations and the master equation)**.** The gauge transformations of the proof of Proposition **??** are, in components,

$$A_m \mapsto D_m c, \qquad A_I \mapsto [A_I, c], \qquad \lambda_\alpha{}^i \mapsto [c, \lambda_\alpha{}^i], \qquad \bar\lambda_{\dot\alpha i} \mapsto [c, \bar\lambda_{\dot\alpha i}], \qquad c \mapsto \tfrac{1}{2}[c, c],$$

the coefficients of A^*_m, A^*_I, $\lambda^{*\alpha}{}_i$, $-\bar\lambda^{*\dot\alpha i}$ and c^* in the action; $\{S, S\} = 0$ is the classical BV master equation (Zinn-Justin equation) and $QI + \frac{1}{2}\{I, I\} = 0$ its interacting part. The functionals of the proof are

$$\int d^4x\,\big(\lambda, \rho([dc, \lambda])\big) = i\int d^4x \Big\{\big(\lambda_\alpha{}^i, \bar\sigma^{m\dot\alpha\alpha}[\partial_m c, \bar\lambda_{\dot\alpha i}]\big) + \big(\bar\lambda_{\dot\alpha i}, \bar\sigma^{m\dot\alpha\alpha}[\partial_m c, \lambda_\alpha{}^i]\big)\Big\},$$

$$\int d^4x\,\big(df, \Gamma(s_1, s_2)\big) = \int d^4x\,\partial_m f\,\Gamma^m(s_1, s_2), \qquad \int d^4x\,([dc, c], A^*) = \int d^4x\,([\partial_m c, c], A^*_m).$$

Chapter 3

Supplements 2

Remark 3.1 (Physics notation: conformal and R-symmetry generators)**.** The even generators are the vector fields $v^m(x)\partial_m$, with $X_{mn} = \partial_m v_n - \partial_n v_m$:

- translation P_b: $v^m = b^m$, so $P_m = \partial_m$;
- rotation with $\omega_{mn} = -\omega_{nm}$: $v^m = \omega_{mn}x_n$, $X_{mn} = -2\omega_{mn}$, the Euclidean Lorentz algebra $\mathfrak{so}(4) = \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$;
- dilatation $\Delta = 2E$: $v^m = 2x^m$, $\operatorname{div} v = 8$;
- special conformal K_b: $v^m = x^2 b^m - 2(b\cdot x)x^m$, $X_{mn} = 4(x_m b_n - x_n b_m)$, $\operatorname{div} v = -8\, b\cdot x$;
- R-symmetry $R \in \mathfrak{so}(6)$: $v(y)^I = R_{IJ}y_J$ on $W^\perp_{\mathbb{R}}$, acting on W by the traceless $r^i{}_j = \frac{1}{4}R_{IJ}(\bar\Sigma^J\Sigma^I)^i{}_j$ with $\bar\Sigma^I = (\Sigma^I)^\dagger$, an element of the complexified $\mathfrak{su}(4)_R$.

The even part is $\mathfrak{sl}(4,\mathbb{C})\oplus\mathfrak{sl}(4,\mathbb{C})$, the complexification of the conformal algebra $\mathfrak{su}(2,2)$ and of $\mathfrak{su}(4)_R$.

Remark 3.2 (Physics notation: conformal Killing spinors)**.** With $Q_1 = (\xi_\alpha{}^i, \bar\xi_{\dot\alpha i})$, the parameters of Q^i_α and $\bar Q_{\dot\alpha i}$, and $Q_2 = (\eta^\alpha{}_i, \bar\eta^{\dot\alpha i})$, the parameters of $S^\alpha{}_i$ and $\bar S^{\dot\alpha i}$, the conformal Killing spinor $x \mapsto Q_1 + \rho(x)Q_2$ is

$$\epsilon_\alpha{}^i(x) = \xi_\alpha{}^i - i\, x_m \sigma^m_{\alpha\dot\alpha}\bar\eta^{\dot\alpha i}, \qquad \bar\epsilon_{\dot\alpha i}(x) = \bar\xi_{\dot\alpha i} + i\, x_m \eta^\alpha{}_i \sigma^m_{\alpha\dot\alpha}:$$

the parameter $\bar\eta$ of $\bar S$ enters the undotted ϵ, and the parameter η of S the dotted $\bar\epsilon$. The components ξ, $\bar\xi$, η, $\bar\eta$ are commuting numbers; the odd parity of the generators sits in Π.

Remark 3.3 (Physics notation: the brackets of the superconformal algebra)**.** For $\epsilon_a = \xi_a + \rho(x)\eta_a$, $a = 1, 2$,

$$[\epsilon_1, \epsilon_2] = \Big(x \mapsto i\big(\bar\epsilon_{1\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\epsilon_{2\alpha}{}^i + \bar\epsilon_{2\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\epsilon_{1\alpha}{}^i\big)\partial_m\Big) + r,$$
$$r^k{}_i = 2\big(\xi_{1\alpha}{}^k\eta_2{}^\alpha{}_i + \bar\xi_{1\dot\alpha i}\bar\eta_2^{\dot\alpha k}\big) + 2\big(\xi_{2\alpha}{}^k\eta_1{}^\alpha{}_i + \bar\xi_{2\dot\alpha i}\bar\eta_1^{\dot\alpha k}\big) - \text{trace},$$

with the rotation of $W^\perp_{\mathbb{R}}$ written as the $SU(4)_R$ rotation $r^k{}_i$ of W. For an even generator with v^m, X_{mn}, $\operatorname{div} v$ and $r^i{}_j$ as after Definition **??**,

$$[v,\epsilon]_\alpha{}^i = v^n\partial_n\epsilon_\alpha{}^i - \tfrac{1}{8}X_{mn}(\sigma^m\bar\sigma^n)_\alpha{}^\beta\epsilon_\beta{}^i - \tfrac{1}{8}\operatorname{div} v\,\epsilon_\alpha{}^i + r^i{}_j\epsilon_\alpha{}^j,$$
$$[v,\epsilon]_{\dot\alpha i} = v^n\partial_n\bar\epsilon_{\dot\alpha i} + \tfrac{1}{8}X_{mn}\bar\epsilon_{\dot\beta i}(\bar\sigma^m\sigma^n)^{\dot\beta}{}_{\dot\alpha} - \tfrac{1}{8}\operatorname{div} v\,\bar\epsilon_{\dot\alpha i} - \bar\epsilon_{\dot\alpha j}r^j{}_i,$$

so $\frac{1}{4}\rho(d(v,-))$ is the spin term $-\frac{1}{8}X_{mn}\sigma^m\bar\sigma^n$ and $-\frac{1}{8}\operatorname{div} v = -1$ for $\Delta = 2E$. For a rotation the spin terms of $[v,\epsilon]_\alpha{}^i$ and $[v,\epsilon]_{\dot\alpha i}$ are

$$\tfrac{1}{4}\omega_{mn}(\sigma^m\bar\sigma^n)_\alpha{}^\beta\epsilon_\beta{}^i, \qquad -\tfrac{1}{4}\omega_{mn}\bar\epsilon_{\dot\beta i}(\bar\sigma^m\sigma^n)^{\dot\beta}{}_{\dot\alpha}.$$

Remark 3.4 (Physics notation: named generators and their brackets)**.** Under $SU(2)_L \times SU(2)_R \times SU(4)_R$ the named subspaces are: P_m and K_m in $(\mathbf{2},\mathbf{2},\mathbf{1})$, the rotations in $(\mathbf{3},\mathbf{1},\mathbf{1}) \oplus (\mathbf{1},\mathbf{3},\mathbf{1})$, Δ a singlet, the R-symmetry in $(\mathbf{1},\mathbf{1},\mathbf{15})$, Q^i_α in $S_+ \otimes W = (\mathbf{2},\mathbf{1},\mathbf{4})$, $\bar{Q}_{\dot\alpha i}$ in $S_- \otimes W^* = (\mathbf{1},\mathbf{2},\bar{\mathbf{4}})$, $\bar{S}^{\dot\alpha i}$ in $S_- \otimes W = (\mathbf{1},\mathbf{2},\mathbf{4})$ and $S^\alpha{}_i$ in $S_+ \otimes W^* = (\mathbf{2},\mathbf{1},\bar{\mathbf{4}})$. In the normalisation $\Delta = -E$ of [SW, (3)] the eigenvalues of ad_Δ are the scaling dimensions 1 of P, $\frac{1}{2}$ of $Q, \bar{Q}$, $-\frac{1}{2}$ of $S, \bar{S}$ and -1 of K. The brackets of Proposition **??** read:

- $Q_1, Q_2 \in \Sigma$: the translation $v^m = i\big(\bar\xi_{2\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\xi_{1\alpha}{}^i + \bar\xi_{1\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\xi_{2\alpha}{}^i\big)$, pairing Q^i_α with $\bar{Q}_{\dot\alpha i}$;
- $Q_1, Q_2 \in \Sigma^*$: $-K_{v_1}$ with $v_1^m = -i\big(\eta_2{}^\alpha{}_i\sigma^m_{\alpha\dot\alpha}\bar\eta_1^{\dot\alpha i} + \eta_1{}^\alpha{}_i\sigma^m_{\alpha\dot\alpha}\bar\eta_2^{\dot\alpha i}\big)$, pairing $S^\alpha{}_i$ with $\bar{S}^{\dot\alpha i}$;
- $Q_1 \in \Sigma$, $Q_2 \in \Sigma^*$: the dilatation part $\frac{1}{2}\big(\xi_\alpha{}^i\eta^\alpha{}_i - \bar\xi_{\dot\alpha i}\bar\eta^{\dot\alpha i}\big)\Delta$, a rotation, and the R-symmetry rotation r;
- translations: $[b,\eta] = \rho(b)\eta$ with $(\rho(b)\eta)_\alpha{}^i = -i\,b_m\sigma^m_{\alpha\dot\alpha}\bar\eta^{\dot\alpha i}$, from $\bar{S}$ to Q, and $(\rho(b)\eta)_{\dot\alpha i} = i\,b_m\eta^\alpha{}_i\sigma^m_{\alpha\dot\alpha}$, from S to $\bar{Q}$;
- special conformal: $[K_b,\xi] = \rho(b)\xi$ with $(\rho(b)\xi)^{\dot\alpha i} = -i\,b_m\bar\sigma^{m\dot\alpha\alpha}\xi_\alpha{}^i$, from Q to $\bar{S}$, and $(\rho(b)\xi)^\alpha{}_i = i\,b_m\bar\xi_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}$, from $\bar{Q}$ to S;
- $[\Delta, Q_1] = -Q_1$ on Q^i_α, $\bar{Q}_{\dot\alpha i}$ and $[\Delta, Q_2] = +Q_2$ on $S^\alpha{}_i$, $\bar{S}^{\dot\alpha i}$, for $\Delta = 2E$.

Remark 3.5 (Physics notation: constant ghosts)**.** A cochain $f(x_1,\dots,x_k)$ is the coefficient of k constant ghosts of $\mathfrak{psl}(4|4)$, one for each argument, all of ghost number $+1$: the ghosts of P_m, K_m, Δ, the rotations and the R-symmetry are Grassmann odd, and those of Q^i_α, $\bar{Q}_{\dot\alpha i}$, $S^\alpha{}_i$, $\bar{S}^{\dot\alpha i}$ are the commuting spinors $\xi_\alpha{}^i$, $\bar\xi_{\dot\alpha i}$, $\eta^\alpha{}_i$, $\bar\eta^{\dot\alpha i}$. An element of degree 0 has total ghost number 0, so $S^{(k)}$ has ghost number $-k$, carried by its antifields. d_{CE} is the BRST differential of the constant ghosts, the two sign conventions of Lemma **??** differ by the sign of every constant ghost, and (**??**) is the classical master equation of $S + S^{(1)} + S^{(2)}$ extended by the constant ghosts.

Remark 3.6 (Physics notation: conformal and R-symmetry transformations of the fields)**.** Every $S^{(1)}(x_1)$ has the form

$$S^{(1)}(x_1) = \int d^4x \Big\{ (x_1.A_m, A^*_m) + \tfrac{1}{4}(x_1.A_{ij}, A^{*ij}) - (x_1.\lambda_\alpha{}^i, \lambda^{*\alpha}{}_i) + (x_1.\bar\lambda_{\dot\alpha i}, \bar\lambda^{*\dot\alpha i}) - (x_1.c, c^*) \Big\},$$

with $x_1.\varphi$ the transformation of the field φ along x_1. For an even generator with v^m, X_{mn}, $\operatorname{div} v$ and $r^i{}_j$ as after Definition **??**,

$$\begin{aligned}
&v.A_m = v^n\partial_n A_m + A_n\partial_m v^n, \qquad v.A_{ij} = v^n\partial_n A_{ij} + \tfrac{1}{4}\operatorname{div} v\, A_{ij} - r^k{}_i A_{kj} - r^k{}_j A_{ik}, \qquad v.c = v^n\partial_n c,\\
&v.\lambda_\alpha{}^i = v^n\partial_n\lambda_\alpha{}^i - \tfrac{1}{8}X_{mn}(\sigma^m\bar\sigma^n)_\alpha{}^\beta\lambda_\beta{}^i + \tfrac{3}{8}\operatorname{div} v\,\lambda_\alpha{}^i + r^i{}_j\lambda_\alpha{}^j,\\
&v.\bar\lambda_{\dot\alpha i} = v^n\partial_n\bar\lambda_{\dot\alpha i} + \tfrac{1}{8}X_{mn}\bar\lambda_{\dot\beta i}(\bar\sigma^m\sigma^n)^{\dot\beta}{}_{\dot\alpha} + \tfrac{3}{8}\operatorname{div} v\,\bar\lambda_{\dot\alpha i} - \bar\lambda_{\dot\alpha j}r^j{}_i :
\end{aligned}$$

the conformal transformations of a vector, of six scalars of Weyl weight 1 and of Weyl spinors of Weyl weight $\frac{3}{2}$, the coefficients $\frac{1}{4}$ and $\frac{3}{8}$ of $\operatorname{div} v$ being the weight divided by 4, together with the $SU(4)_R$ rotation r.

Remark 3.7 (Physics notation: supersymmetry and special superconformal transformations)**.** For $\xi \in \Sigma$, (**??**) gives $\xi.A = -\Gamma(\xi,\lambda)$ and $\xi.\lambda = -\frac{1}{2}\rho(F_A)\xi$, in components

$$\begin{aligned}
&\xi.A_m = -i\big(\bar\lambda_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\xi_\alpha{}^i + \bar\xi_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\lambda_\alpha{}^i\big), \qquad \xi.A_{kl} = -2\varepsilon_{jikl}\xi_\alpha{}^j\varepsilon^{\alpha\beta}\lambda_\beta{}^i + 2\big(\bar\xi_{\dot\alpha k}\varepsilon^{\dot\alpha\dot\beta}\bar\lambda_{\dot\beta l} - \bar\xi_{\dot\alpha l}\varepsilon^{\dot\alpha\dot\beta}\bar\lambda_{\dot\beta k}\big),\\
&\xi.\lambda_\alpha{}^i = \tfrac{1}{4}F_{mn}(\sigma^m\bar\sigma^n)_\alpha{}^\beta\xi_\beta{}^i + \tfrac{1}{4}[A^{ik}, A_{kj}]\,\xi_\alpha{}^j + \tfrac{i}{2}\,\sigma^m_{\alpha\dot\alpha}\varepsilon^{\dot\alpha\dot\beta}D_m A^{ij}\,\bar\xi_{\dot\beta j},\\
&\xi.\bar\lambda_{\dot\alpha i} = -\tfrac{1}{4}F_{mn}\bar\xi_{\dot\beta i}(\bar\sigma^m\sigma^n)^{\dot\beta}{}_{\dot\alpha} + \tfrac{1}{4}[A_{ik}, A^{kj}]\,\bar\xi_{\dot\alpha j} - \tfrac{i}{2}\,\varepsilon^{\alpha\beta}\sigma^m_{\alpha\dot\alpha}D_m A_{ij}\,\xi_\beta{}^j,
\end{aligned}$$

the supersymmetry transformations along Q^i_α and $\bar{Q}_{\dot\alpha i}$ with commuting parameters, so that $\xi.A_m$ and $\xi.A_{kl}$ are Grassmann odd. For $\eta \in \Sigma^*$, (**??**) gives the special superconformal transformations: the same formulas with $(\xi_\alpha{}^i, \bar\xi_{\dot\alpha i})$ replaced by $(\epsilon_\alpha{}^i(x), \bar\epsilon_{\dot\alpha i}(x))$, and in $\eta.\lambda$ the scalar terms $\rho(A|_{W^\perp_{\mathbb{R}}})\eta$,

$$\eta.\lambda_\alpha{}^k \supset -A^{kj}\varepsilon_{\alpha\beta}\eta^\beta{}_j, \qquad \eta.\bar\lambda_{\dot\beta k} \supset -A_{kj}\varepsilon_{\dot\beta\dot\alpha}\bar\eta^{\dot\alpha j}.$$

In Example **??**, $\Delta.A_M = 2x^m\partial_m A_M + 2A_M$, $\Delta.\lambda = 2x^m\partial_m\lambda + 3\lambda$, $\Delta.c = 2x^m\partial_m c$. In Proposition **??** the spin coefficient is that of $-\frac{1}{8}X_{mn}\sigma^m\bar\sigma^n$, the two Weyl coefficients are those of $\frac{3}{8}\operatorname{div} v$ on λ and $\frac{1}{4}\operatorname{div} v$ on A_I, and the coefficient -1 is that of the scalar terms above.

Remark 3.8 (Physics notation: the quadratic-antifield term)**.** For ϵ_1, ϵ_2 each a constant ξ or $\rho(x)\eta$,

$$S^{(2)}(\epsilon_1,\epsilon_2) = \int d^4x \Big\{ \tfrac{1}{4}\Big[-2i\,\Gamma^m(\epsilon_1,\epsilon_2)\big(\lambda^{*\alpha}{}_i, \sigma^m_{\alpha\dot\alpha}\bar\lambda^{*\dot\alpha i}\big) + \tfrac{1}{4}\Gamma_{kl}(\epsilon_1,\epsilon_2)\Gamma^{kl}(\lambda^*,\lambda^*)\Big] - \tfrac{1}{2}\big((\epsilon_1,\lambda^*),(\epsilon_2,\lambda^*)\big) \\ - \big(\Gamma^m(\epsilon_1,\epsilon_2)A_m + \tfrac{1}{4}\Gamma_{ij}(\epsilon_1,\epsilon_2)A^{ij},\, c^*\big)\Big\},$$

with Γ^m, Γ_{kl} the components of the Γ-pairing on Σ evaluated on $(\epsilon_\alpha{}^i(x), \bar\epsilon_{\dot\alpha i}(x))$, $(\epsilon,\lambda^*) = \epsilon_\alpha{}^i\lambda^{*\alpha}{}_i - \bar\epsilon_{\dot\alpha i}\bar\lambda^{*\dot\alpha i}$ and

$$\Gamma^{kl}(\lambda^*,\lambda^*) = -2\varepsilon^{klpj}\big(\varepsilon_{\alpha\beta}\lambda^{*\beta}{}_j, \lambda^{*\alpha}{}_p\big) + 2\big(\varepsilon_{\dot\beta\dot\alpha}\bar\lambda^{*\dot\alpha l}, \bar\lambda^{*\dot\beta k}\big) - 2\big(\varepsilon_{\dot\beta\dot\alpha}\bar\lambda^{*\dot\alpha k}, \bar\lambda^{*\dot\beta l}\big).$$

In $\{S, S^{(2)}\}$ the $\lambda^*\lambda^*$ terms produce the gaugino equation of motion, and the c^* term the gauge transformation with the field-dependent parameter $\Gamma^m A_m + \frac{1}{4}\Gamma_{ij}A^{ij} = \Gamma^m A_m + \Gamma^I A_I$.

Remark 3.9 (Physics notation: gauge parameters and the end of the tower)**.** The gauge parameter $\iota_{\Gamma(Q_1,Q_2)}A = \Gamma^m A_m + \Gamma^I A_I$ is:

- $Q_1, Q_2 \in \Sigma$: constant, with $\Gamma^m = i\big(\bar\xi_{2\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\xi_{1\alpha}{}^i + \bar\xi_{1\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\xi_{2\alpha}{}^i\big)$ and $\Gamma_{kl}(\xi_1,\xi_2)$;
- $Q_1 \in \Sigma$, $Q_2 \in \Sigma^*$: the same components evaluated on ξ_1 and on $\rho(x)\eta_2$, linear in x;
- $Q_1, Q_2 \in \Sigma^*$: $-\big(x^2 v_1^m - 2(v_1\cdot x)x^m\big)A_m - \frac{1}{4}x^2\,\Gamma^{kl}(\eta_1,\eta_2)A_{kl}$, with $v_1^m = -i\big(\eta_2{}^\alpha{}_i\sigma^m_{\alpha\dot\alpha}\bar\eta_1^{\dot\alpha i} + \eta_1{}^\alpha{}_i\sigma^m_{\alpha\dot\alpha}\bar\eta_2^{\dot\alpha i}\big)$.

$S^{(2)}$ is built from A_m, A_I, $\lambda^{*\alpha}{}_i$, $\bar\lambda^{*\dot\alpha i}$ and c^*, which pair with A^*_M, $\lambda_\alpha{}^i$, $\bar\lambda_{\dot\alpha i}$ and c, so $\{S^{(2)}, S^{(2)}\} = 0$; with $S^{(k\geq 3)} = 0$ the equivariant action is at most quadratic in the constant ghosts.

Remark 3.10 (Physics notation: closure and the ten-dimensional input)**.** The subalgebra of Axiom **??**, generated by P_m, the rotations, the R-symmetry, Q^i_α and $\bar{Q}_{\dot\alpha i}$, is the $\mathcal{N} = 4$ super-Poincaré algebra with its R-symmetry, obtained by reduction from ten-dimensional super Yang–Mills theory with gaugino in the **16** of Spin(10). In the physical reading, for abelian free fields with $\xi.A = -\Gamma(\xi,\lambda)$, $\xi.\lambda = -\frac{1}{2}\rho(F_A)\xi$ and $v = \Gamma(\xi_a,\xi_b)$,

$$\{\xi_a., \xi_b.\}A_m = -v^n F_{nm} + v^I\partial_m A_I = -v^n\partial_n A_m + \partial_m\big(v^n A_n + v^I A_I\big), \qquad \{\xi_a., \xi_b.\}A_{ij} = -v^n\partial_n A_{ij},$$

the translation $-v^n\partial_n$ together with the gauge transformation with parameter $\iota_v A = v^n A_n + v^I A_I$ of the c^* term of $S^{(2)}$. In Lemma **??** the tree tensors are, for $\mathfrak{su}(N)$ and up to normalisation, products of structure constants f^{abc} contracted with δ^{ab} along a tree, and δ^{ab} for $k = 2$.